1)

$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$

$P(X \geq 7) = P(X=7) + P(X=8) + P(X=9) + P(X=10)$

$P(X \geq 7) = \left(\binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10} \right) \times 0.5^{10}$

$= (120 + 45 + 10 + 1) \times 0.5^{10} = 0.171875$

2.1)

$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$

$P(X=5) = \binom{5}{5} \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^0 = 0.1317$

2.2)

$P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$

$P(X=3) = \binom{5}{3} \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^2 = 10 \times \frac{8}{27} \times \frac{1}{9} = \frac{80}{243}$

$P(X=4) = \binom{5}{4} \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^1 = 5 \times \frac{16}{81} \times \frac{1}{3} = \frac{80}{243}$

$P(X=5) = \binom{5}{5} \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^0 = 1 \times \frac{32}{243} \times 1 = \frac{32}{243}$

$P(X \geq 3) = \frac{80}{243} + \frac{80}{243} + \frac{32}{243} = \frac{192}{243} = 0.7901$

2.3)

$P(X=2) = \binom{5}{2} \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^3 = 10 \times \frac{4}{9} \times \frac{1}{3} = \frac{40}{243} = 0.1646$

3)

$P(X=x) = p\,(1-p)^{n-1}$

$p = \frac{3}{75} = \frac{1}{25} = 0.04$

$P(X=6) = 0.04 \times (0.96)^5 =$ 0.3261

4.1)

$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

$\lambda = 1$

$P(X=0) = e^{-1} =$ 0.3679

4.2)

$P(X=1) = e^{-1} =$ 0.3679

4.3)

$P(X=2) = \frac{e^{-1} \cdot 1^2}{2} = \frac{e^{-1}}{2} =$ 0.1839

4.4)

$P(X=3) = \frac{e^{-1} \cdot 1^3}{3!} = \frac{e^{-1}}{6} =$ 0.0613